Momin M. Malik, PhD

critical computational methodology • data science & society • network modeling

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STATEMENT OF PURPOSE

I bring statistics and machine learning together with critical perspectives from social science to consider when, how, and why data and modeling succeed in their aims—and when, how, and why they can fail. I am passionate about improving practice towards more responsible, robust, effective, and just uses of data and modeling, as well as engaging in outreach to help practitioners in policy, public health, social work, law, government, journalism, social science, industry, civil society, and elsewhere understand and adopt machine learning and data science.

EDUCATION

School of Computer Science, Carnegie Mellon University, Aug 2013–Aug 2018
PhD in Societal Computing and MS in Computation, Organizations and Society (Institute for Software Research) and MS in Machine Learning (Machine Learning Department). Research on social media and sensor data. ARCS Foundation award. Dissertation: "Bias and beyond in digital trace data." Committee: Jürgen Pfeffer and Anind K. Dey (Coadvisors), Cosma R. Shalizi (Department of Statistics), and David Lazer (Northeastern University).

Oxford Internet Institute, University of Oxford, Oct 2011–Sep 2012
MSc with distinction in Social Science of the Internet. Master's thesis on locating the emergence of Internet studies in the 20-year evolution of a large co-authorship network. Advisor: Eric T. Meyer.

Department of the History of Science, Harvard University, Sep 2004–Mar 2009
AB cum laude in History and Science with Music minor. Senior thesis on narratives of the early 20th century South Indian mathematician Srinivasa Ramanujan. Advisor: Lukas Rieppel.

WORK AND RESEARCH EXPERIENCE

Director of Data Science, Avant-garde Health. Boston, MA, Nov 2020–present
Leading a small team of data scientists at a healthcare data startup, born out of Harvard Business School's value-based healthcare research and serving about a dozen client hospitals. Research into value-based healthcare; predictive modeling to support bundling payments for Medicare and Medicaid patients; integrating analysis techniques into the product platform.

Data Analyst / Scientist (Temp 2020), MoveOn. Remote, Aug 2020–Nov 2020
Analytics support for progressive organizing efforts around the US 2020 General Election. Tracking disinformation and digital voter suppression targeting Black and minority voters; designing and implementing voter mobilization experiments around ballot drop box locations.

Data Science Postdoctoral Fellow, Berkman Klein Center for Internet & Society at Harvard University. Cambridge, MA, Sep 2018–Aug 2020
Statistical learning and network modeling for data in Media Cloud and other projects; technical advising and tutoring for scholars and practitioners in social science, journalism, law, and policy; building in-house data science capacity; forging stronger university-wide ties between social and data science; and research into conceptual, ethical, theoretical, and practical challenges of deploying data science.

Data Science for Social Good Fellow. Lisbon, Portugal, Summer 2017
Project for Tuscan agencies applying machine learning and network analysis to urban data for sustainable tourism in Florence. Run by the Center for Data Science and Public Policy, University of Chicago, and Nova School of Business and Economics, Universidade NOVA de Lisboa.

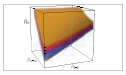
Graduate research assistant, Ubicomp Lab. Pittsburgh, PA, Jun 2016–Aug 2018
Research under Dr. Anind Dey (Human-Computer Interaction Institute and University of Washington) on mobile phone sensor collection of social network data. Ran 3-month 53-subject study, collecting mobile phone sensor data alongside social network survey responses.

Graduate research assistant, Pfeffer Lab. Pittsburgh, PA, Aug 2013–Jun 2018
Research under Dr. Jürgen Pfeffer (Institute for Software Research and Technical University of Munich) on biases in social media data (representativeness, data access, effect of platform constraints on behavior), and news media and social media. Use of Twitter 10% sample.

Research assistant, *Big Data: A Revolution That Will Transform How We Live, Work, and Think*. Oxford, UK, 2011–2012
Research for 2013 book by Prof. Viktor Mayer-Schönberger (Oxford Internet Institute) and Kenneth Cukier (*The Economist*). History of statistics and data management, historical and modern case studies, fact-checking, idea development, and editing draft material.

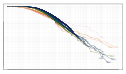
Research assistant, Berkman Center for Internet & Society at Harvard University. Cambridge, MA, 2009–2011 & 2012–2013
Research for Dr. Urs Gasser (Executive Director, Berkman Klein Center) on 'Youth and Media' project. Multidisciplinary review of youth information-seeking; policy reports; focus group design; analysis of qualitative data on youth social media usage; theory of 'news literacy.'

PUBLICATIONS

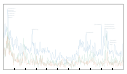


Eugene T. Richardson, **Momin M. Malik**, William A. Darity, Jr., A. Kirsten Mullen, Michelle E. Morse, Maya Malik, Aletha Maybank, Mary T. Bassett, Paul E. Farmer, Lee Worden, and James Holland Jones. 2021. Reparations for American descendants of persons enslaved in the U.S. and their potential impact on SARS-CoV-2 transmission. *Social Science & Medicine*.

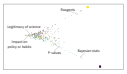
PUBLICATIONS, continued



Hal Roberts, Rahul Bhargava, Linas Valiukas, Dennis Jen, **Momin M. Malik**, Cindy Bishop, Emily Ndulue, Aashka Dave, Anushka Shah, Jasmin Rubinowitz, Justin Clark, Bruce Etling, Rob Faris, Alexis Hope, Catherine D'Ignazio, Fernando Bermejo, Yochai Benkler, and Ethan Zuckerman. 2021 (forthcoming). Media Cloud: Massive open source collection of global news on the open web. In *Proceedings of the 15th International Conference on Web and Social Media* (ICWSM-2021).

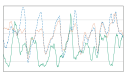


Mutale Nkonde, Maria Y. Rodriguez, Leo Cortana, Joan K. Mukogosi, Shakira King, Ray Serrato, Natalie Martinez, Mary Drummer, Ann Lewis, and **Momin M. Malik**. 2021. Disinformation creep: ADOS and the weaponization of breaking news. *Harvard Kennedy School (HKS) Misinformation Review* 1 (7).

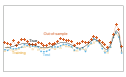


Nicole C. Nelson, Kelsey Ichikawa, Julie Chung, and **Momin M. Malik**. 2021 (in submission). Mapping the discursive dimensions of the reproducibility crisis: A mixed methods analysis. MetaArXiv:10.31222/osf.io/sbv3q.

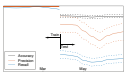
Momin M. Malik and Maya Malik. 2021 (in submission). Critical technical awakenings.



Momin M. Malik, Afsaneh Doryab, Mike A. Merrill, Jürgen Pfeffer, and Anind K. Dey. 2021 (in submission). Can co-location data detect friendship? It depends how you model it. arXiv: 2008.02919

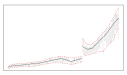


Momin M. Malik. 2020 (in submission). A hierarchy of limitations in machine learning. arXiv: 2002.05193.

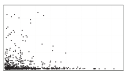


Kar-Hai Chu, Jason Colditz, **Momin M. Malik**, Tabitha Yates, and Brian Primack. 2019. Identifying key target audiences for public health campaigns: Leveraging machine learning in the case of hookah tobacco smoking. *Journal of Medical Internet Research* 21 (7): e12443.

Jürgen Pfeffer and **Momin M. Malik**. 2017. Simulating the dynamics of socio-economic systems. In *Networked governance: New research perspectives*, edited by Betina Hollstein, Wenzel Matiaske, and Kai-Uwe Schnapp, 143–161. Cham, Switzerland: Springer.

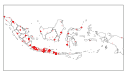


Momin M. Malik and Jürgen Pfeffer. 2016. Identifying platform effects in social media data. In *Proceedings of the Tenth International AAAI Conference on Web and Social Media* (ICWSM-16), 241-249.

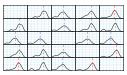


Momin M. Malik and Jürgen Pfeffer. 2016. A macroscopic analysis of news content on Twitter. *Digital Journalism* 4 (8): 955-979.

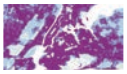
Gabriel Ferreira, **Momin Malik**, Christian Kästner, Jürgen Pfeffer, and Sven Apel. 2016. Do #ifdefs influence the occurrence of vulnerabilities? An empirical study of the Linux Kernel. In *Proceedings of the 20th International Systems and Software Product Line Conference* (SPLC '16), 65-73.



Kathleen M. Carley, **Momin Malik**, Peter M. Landwehr, Jürgen Pfeffer, and Michael Kowalchuck. 2016. Crowd sourcing disaster management: The complex nature of Twitter usage in Padang Indonesia. *Safety Science* 90: 48-61.



Hemank Lamba, **Momin M. Malik**, and Jürgen Pfeffer. 2015. A tempest in a teacup? Analyzing firestorms on Twitter. In *Proceedings of the 2015 Conference on Advances in Social Networks Analysis and Mining* (ASONAM 2015), 17-24. **Best Student Paper Award**.



Momin M. Malik, Hemank Lamba, Constantine Nakos, and Jürgen Pfeffer. 2015. Population bias in geotagged tweets. In *Papers from the 2015 ICWSM Workshop on Standards and Practices in Large-Scale Social Media Research* (ICWSM-15 SPSM), 18-27.

REPORTS

Io Flament, Cristina Lozano, and **Momin M. Malik**. (2017). *Data-driven planning for sustainable tourism in Tuscany.* Cascais, Portugal: Data Science for Social Good Europe.

Momin Malik, Sandra Cortesi, and Urs Gasser. (2013, October 18). *The challenges of defining 'news literacy'.* Berkman Center Research Publication No. 2013-20.

Urs Gasser, Sandra Cortesi, **Momin Malik**, and Ashley Lee. (2012, February 16). *Youth and digital media: From credibility to information quality.* Berkman Center Research Publication No. 2012-1.

AFFILIATIONS

Affiliate, Berkman Klein Center for Internet & Society at Harvard Univeristy, 2020–present.

Fellow, Institute in Critical Quantitative, Computational, & Mixed Methodologies (ICQCM), 2021–present.

BLOGGING

"Can algorithms themselves be biased?" *Medium,* Berkman Klein Center Collection. April 24, 2019. https://link.medium.com/ULLHZJQL3Y.

TEACHING

"A critical introduction to machine learning." 2019 ACM Richard Tapia Celebration of Diversity in Computing Conference (Tapia 2019). September 19, 2019, Marriott Marquis San Diego Marina, San Diego, California.

"Everything you ever wanted to know about network statistics but were afraid to ask." 3-hour tutorial. XXXIX Sunbelt Social Networks Conference of the International Network for Social Network Analysis (Sunbelt 2019). June 18, 2019, Montreal, Quebec.

Future Faculty Program (completed), Eberly Center for Teaching Excellence and Educational Innovation, Jan 2014 – Mar 2018
Enrolled since first year of PhD. Two teaching feedback consultations. Nine seminars on topics like student motivation and engagement, discussion facilitation, lecture design and delivery, course and syllabus design, teaching effectiveness monitoring.

Teaching Assistant, Ethics and Policy Issues in Computing (Instructor: Dr. Jim Herbsleb), Spring 2016
Graded weekly response papers and activity on online class forum. Gave a lecture overview of rhetoric and debate in preparation for assigned one-on-one debates. Provided template and gave design tutorial for scientific poster making. Helped students with course projects (scope, references) and posters. Guest lecture: "Ethical and policy issues in predictive modeling."

Teaching Assistant, Introduction to Network Science (Instructor: Dr. Jürgen Pfeffer), Fall 2015
Wrote four comprehensive solution sets to problems from Mark E. J. Newman's *Networks: An Introduction* (OUP, 2010). Guest lecture tutorial on R and the igraph library for network analysis. Graded problem sets and tests.

PRESENTATIONS

"Critical theory and quantification" (invited session). With Ezekiel Dixon-Román and Maya Malik. Seminar Virtual Event, Histories of Artificial Intelligence: A Genealogy of Power (Mellon Sawyer Seminar series), University of Cambridge, January 20, 2020 [held online].

"A hierarchy of limitations in machine learning." Math and Democracy Seminar Series, Center for Data Science, New York University. October 5, 2020, New York, New York [delivered online].

"A hierarchy of limitations in machine learning: Data biases and the social sciences" (invited webinar). Webinar Series: Data Cultures in Higher Education, Faculty of Psychology and Education, Universitat Oberta de Catalunya (Open University of Catalonia). September 29, 2020, Barcelona, Spain [delivered online].

"Machine learning won't save us: Dependencies bias crossvalidation estimates of model performance." 2020 Virtual Sunbelt Conference of the International Network for Social Network Analysis. July 17, 2020.

"Anti-racism and COVID-19" (invited talk). With Eugene T. Richardson, William A. Darity, Jr., James Holland Jones, A. Kirsten Mullen, and Paul E. Farmer. Global Health and Social Medicine Seminar Series, Department of Global Health & Social Medicine, Harvard Medical School. June 3, 2020, Cambridge, Massachusetts [delivered online].

"Media Cloud and media ecosystem research." Social Science One Researcher Onboarding Training Session lightning talk. May 27, 2020.

"Antiracism and COVID-19" (invited talk). With Eugene T. Richardson. Antiracism & Technology Design Seminar, Space Enabled research group, MIT Media Lab. May 13, 2020, Cambridge, Massachusetts [delivered online].

"Critical technical practice revisited: Towards `analytic actors' in data science" (invited talk). STS Circle, Program on Science, Technology & Society, Harvard Kennedy School. March 5, 2020, Cambridge, Massachusetts.

"Revisiting 'all models are wrong': Addressing limitations in big data, machine learning, and computational social science" (invited talk). Wednesdays@NICO Seminar Speaker Series, Northwestern Institute for Complex Systems, Northwestern University. February 5, 2020, Evanston, Illinois.

"How STS can improve data science" (invited talk). STS Lunch Seminar Series, Tufts University. January 27, 2020, Medford, Massachusetts.

"A hierarchy of limitations in machine learning" (invited talk). Microsoft Research New England. December 3, 2019, Cambridge, Massachusetts.

Correlates of oppression: Machine learning and society. Guest lecture in CMS.701/CMS.901: Current Debates in Media, Fall 2019 (Instructor: Dr. Sasha Costanza-Chock). Comparative Media Studies, MIT. October 30, 2019, Cambridge, Massachusetts.

"Statistics and machine learning: Foundations, limitations, and ethics" (invited talk). Colby College Department of Mathematics and Statistics, Colloquium Fall 2019, Colby College. October 7, 2019, Waterville, Maine.

"Three open problems for historians of AI." Towards a History of Artificial Intelligence. May 24, 2019, Columbia University, New York, New York.

"Interpretability is a red herring: Grappling with 'prediction policy problems.' 17th Annual Information Ethics Roundtable: Justice and Fairness in Data Use and Machine Learning. April 5, 2019, Northeastern University, Boston, Massachusetts.

"What can AI do with copyrighted data?" (invited talk). Bracing for Impact – The Artificial Intelligence Challenge: A Roadmap for AI Governance in Canada. Part II: Data, Policy & Innovation. IP Osgoode, Osgoode Hall Law School, York University. March 21, 2019, Toronto

PRESENTATIONS, continued

Reference Library, Toronto, Canada.

"The ethical implications of technical limitations" (invited talk). Fairness, Accountability & Transparency/Asia. Digital Asia Hub and ACM/FAT*. January 12, 2019, Shun Hing College, University of Hong Kong, Hong Kong.

"Machine learning for social scientists." Fairness, Accountability & Transparency/Asia. Digital Asia Hub and ACM/FAT*. January 11, 2019, Shun Hing College, University of Hong Kong, Hong Kong.

"'AI' is a lie: Getting to the real issues." AGTech Forum, Berkman Klein Center for Internet & Society at Harvard University. December 13, 2018, Cambridge, Massachusetts.

"Theorizing sensors for social network research." Computational Social Science Institute, UMass Amherst. December 7, 2018, Amherst, Massachusetts.

"What everyone needs to know about 'prediction' in machine learning." Leverhulme Centre for the Future of Intelligence, University of Cambridge. December 3, 2018, Cambridge, UK.

"Anxiety, crisis, and a computational future for journalism." Philip Merrill College of Journalism / College of Information Studies, University of Maryland. November 27, 2018, College Park, Maryland.

"Networks, yeah! The representation of relations." Data & Donuts, digital HKS, Harvard Kennedy School. November 2, 2018, Cambridge, Massachusetts.

"Demystifying AI: Terms of disservice." AI Working Group, Berkman Klein Center for Internet & Society. October 28, 2018, Cambridge, Massachusetts.

"Surprising aspects of 'prediction' in data science." 0213eight, Harvard Alumni Association. October 13, 2018, Cambridge, Massachusetts.

"From the forest to the swamp: Modeling vs. implementation in data science." Techtopia @ Harvard University. October 2, 2018, Cambridge, Massachusetts.

Thesis defense: "Bias and beyond in digital trace data." Institute for Software Research, School of Computer Science, Carnegie Mellon University. August 9, 2018, Pittsburgh, Pennsylvania.

"Friendship and proximity in a fraternity cohort with mobile phone sensors." XXXVIII Sunbelt Conference of the International Network for Social Network Analysis. Modeling network dynamics (ses15.05). July 1, 2018, Utrecht, Netherlands.

"A critical introduction to statistics and machine learning." Cascais Data Science for Social Good Europe Fellowship, Nova School of Business and Economics, Universidade NOVA de Lisboa. August 15, 2017, Lisbon, Portugal.

"A social scientist's guide to network statistics." Guest lecture in 70/73-449: Social, Economic and Information Networks, Fall 2016 (Instructor: Dr. Katharine Anderson). Undergraduate Economics, Carnegie Mellon University. November 10, 2016, Pittsburgh, Pennsylvania.

"Bias busters @ university workshop: A Carnegie Mellon/Google collaboration to address unconscious bias." With Carol Frieze (Carnegie Mellon University, School of Computer Science), Gerry Katilius (Google Pittsburgh), and Diana Marculescu (Carnegie Mellon University, College of Engineering). 2016 ACM Richard Tapia Celebration of Diversity in Computing Conference (Tapia 2016). September 15, 2016, Austin, Texas.

"Platform effects in social media networks." 2nd Annual International Conference on Computational Social Science (IC2S2). Social Networks 1. June 24, 2016, Evanston, Illinois.

"Bias busting @ university workshop – A Carnegie Mellon/Google collaboration to address unconscious bias." With Carol Frieze (Carnegie Mellon University, School of Computer Science), Gerry Katilius (Google Pittsburgh), and Diana Marculescu (Carnegie Mellon University, College of Engineering). Women in Engineering ProActive Network Change Leader Forum (WEPAN 2016). June 15, 2016, Broomfield, Colorado.

"Identifying platform effects in social media data." Tenth International AAAI Conference on Web and Social Media (ICWSM-16). Session I: Biases and Inequalities. May 18, 2016, Cologne, Germany.

"Social media data and computational models of mobility: A review for demography." 2016 ICWSM Workshop on Social Media and Demographic Research (ICWSM-16 SMDR). May 17, 2016, Cologne, Germany.

"Ethical and policy issues in predictive modeling." Guest lecture in 08-200/08-630/19-211: Ethics and Policy Issues in Computing, Spring 2016 (Instructor: Dr. James Herbsleb). School of Computer Science, Carnegie Mellon University. March 1, 2016, Pittsburgh, Pennsylvania.

"Platform effects in social media networks." XXXVI Sunbelt Conference of the International Network for Social Network Analysis. Social Media Networks: Challenges and Solutions (Sunday AM2). April 10, 2016, Newport Beach, California.

"Population bias in geotagged tweets." 2015 ICWSM Workshop on Standards and Practices in Social Media Research (ICWSM-15 SPSM).

PRESENTATIONS, continued

Ninth International AAAI Conference on Web and Social Media (ICWSM-15). May 26, 2015, Oxford, UK.

"Inferring social networks from sensor data." XXXIV Sunbelt Conference of the International Network for Social Network Analysis. Network Data Collection (Saturday AM2). February 22, 2014, St Pete Beach, Florida.

ORGANIZING

Momin M. Malik, Katja Mayer, Hemank Lamba, and Claudia Müller-Brin (co-organizers). Workshop on critical data science. *The 13th International Conference on Web and Social Media* (ICWSM-19). Munich, Germany, June 11, 2019.

Logistics, communication, materials, and tabling for "Train the Trainer workshop," by Bias Busters @ Carnegie Mellon University. Google Pittsburgh, November 1 – 2, 2016. Pittsburgh, Pennsylvania.

VOLUNTEERING/COMMUNITY

Facilitator for Bias Busters @ CMU, 2015-2018
Part of 3-person teams leading inclusivity programming for student/faculty/staff groups of 20-30. Led by Dr. Carol Frieze (director, SCS4ALL and Women@SCS), and created in collaboration with Google Pittsburgh.
Senior program committee, International Conference on Web and Social Media (ICWSM), 2020 – present.

Committee Service

Sponsorship chair, 14th International AAAI Conference on Web and Social Media (ICWSM-2020). June 8 – June 11, 2020. Atlanta, Georgia.

Steering committee, Annotation and Benchmarking on Understanding and Transparency of Machine learning Lifecycles (ABOUT ML). The Partnership on AI (PAI). 2019-2020.

Posters co-chair, 11th International ACM Web Science Conference 2019 (WebSci '19). June 30 – July 3, 2019. Boston, Pennsylvania.

Program committee, International Conference on Web and Social Media (ICWSM), 2018 – 2019.

Program committee, International World Wide Web Conference (WWW/TheWebConf) Web and Society track, 2018 – present.

Program committee, International Conference on Computational Social Science (IC2S2), 2017 – present.

Other participation

Leading thinker session on "Life in the Age of Algorithms." Harvard Graduate School of Education. November 7, 2019, Cambridge, MA. Reported about in: Alison J. Head, Barbara Fister, and Margy MacMillan, (2020, January 15), Information literacy in the age of algorithms: Student experiences with news and information, and the need for change, Project Information Research Institute.

International Expert Workshop on Automated Decision-Making. Center for Information Technology, Society, and Law (ITSL), University of Zurich. September 12–14, 2019, Lavin, Switzerland.

Peer Review

Communications of the ACM (Association for Computing Machinery), 2021.

Sociological Methods & Research (SAGE Journals), 2020.

Big Data & Society (SAGE Journals), 2020.

Editorial Board member for special issue on "Critical Data and Algorithms Studies." *Frontiers in Big Data*, Data Mining and Management (Frontiers Media S.A.), 2019.

Frontiers in Big Data, Data Mining and Management (Frontiers Media S.A.), 2019 – 2020.

EPJ Data Science (Springer), 2018 – 2020.

Digital Journalism (Taylor & Francis), 2016 – 2019.

Journal of Medical Internet Research (JMIR Publications), 2017 – 2018.

LANGUAGES

English. Spoken: Native. Written: Native.
Urdu. Spoken: Fluent. Written: Beginner.
Japanese. Spoken: Conversational. Written: Beginner.

SKILLS

R, igraph, Python, pandas, scikitlearn, MATLAB, Postgres/SQL, Gephi, Pajek, LaTeX, Photoshop, Illustrator, InDesign